PROJECT TITLE -

Federated Traffic-Flow Prediction
with
Communication-Cost Evaluation

TEAM NAME -

MOSAIC

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Problem Description

Urban traffic conditions are highly dynamic, making it difficult to accurately predict near-future traffic and provide efficient routes, particularly for emergency vehicles such as ambulances, fire brigades and police vehicles. Although traffic data is continuously generated through cameras, sensors, GPS and mobile-based mobility sources, centralizing this data can create high communication requirements and privacy concerns. Our problem focuses on using Federated Learning for traffic-flow prediction while evaluating and reducing the communication cost of distributed model training, with Chandigarh and its surrounding corridors, including Zirakpur-side entry and exit routes, as the primary case study.

Key issues

- **Emergency response delays:** Ambulances, fire brigades, police vehicles and other emergency responders can lose critical time in congested road networks.
- **Dynamic traffic conditions:** Traffic flow changes continuously with peak hours, incidents, weather, road closures and special events, making static traffic planning insufficient.
- **Lack of proactive prediction:** Conventional navigation systems primarily react to existing traffic conditions; predicting near-future traffic can enable earlier route decisions.
- **Distributed traffic data:** Traffic information is generated across different roads and junctions through cameras, road sensors, GPS/connected vehicles and other mobility-data sources.
- **Data privacy and transfer constraints:** Sending all locally collected traffic data to a central location can create large communication requirements and may raise privacy and data-governance concerns.
- **Communication overhead in distributed learning:** Federated learning avoids transferring raw local datasets, but repeated exchange of model updates can itself consume significant bandwidth.
- **Congestion and fuel consumption:** Better prediction and routing can help reduce unnecessary delays, stop-and-go movement and associated fuel consumption.
- **Scalability:** A practical solution should be capable of extending beyond Chandigarh to other cities and road networks without requiring all raw traffic data to be centralized.

WORKING STEPS OF THE PROPOSED SOLUTION

1. Traffic Data Collection

Collect traffic information from different **junctions and road segments** using available sources such as cameras, sensors, GPS/connected vehicles, mobile-derived mobility data, and historical traffic data.

2. Local Training

The collected data is processed at its respective location and used to **train a local traffic-prediction model**. The model learns the traffic patterns specific to that location.

3. Model Updates

After local training, the location generates a **model update** containing what the model has learned. **Raw traffic data is not sent** to the central system.

4. Central Server

The model updates from different participating locations are sent to the **central server**, where they are collected for federated aggregation.

5. Global Model

The central server combines the received updates using a federated aggregation method such as **FedAvg** and creates an improved **global traffic-prediction model**.

6. Local Training with Global Model

The updated global model is sent back to the participating locations. Each location uses the global model along with its **local traffic data** for further training.

7. Repeat the Federated Learning Cycle

This process is repeated:

Local Training → Model Update → Central Server → Global Model → Local Training

With every round, the global model can learn from traffic patterns across the participating locations.

8. Traffic Prediction

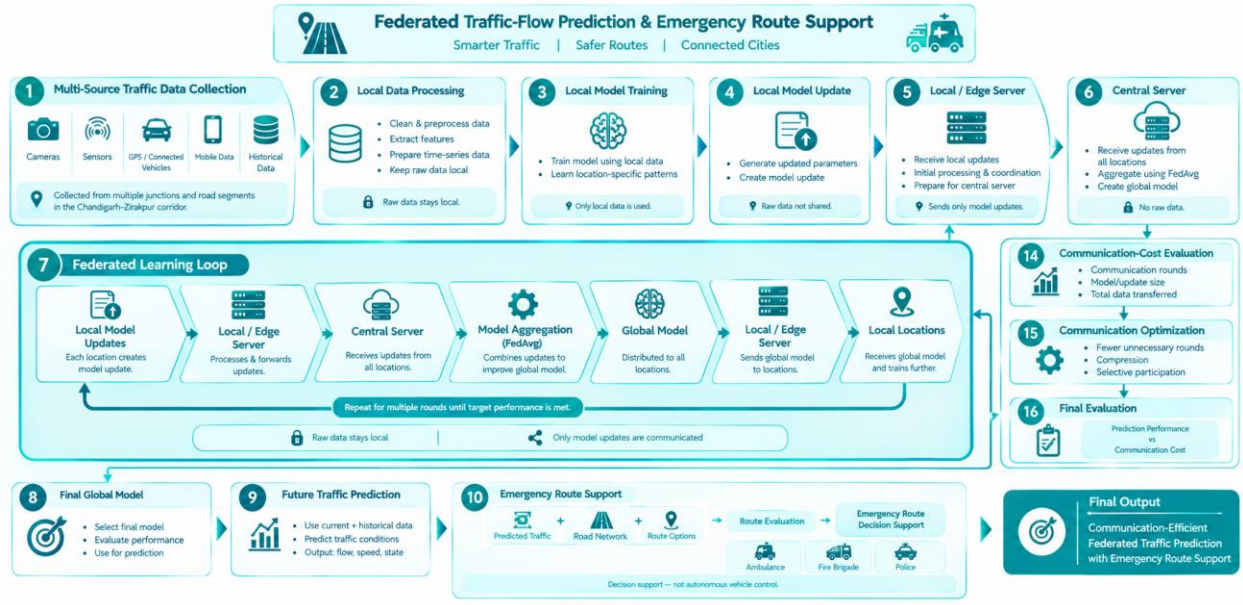
Once the model reaches the required performance, the final global model is used to **predict future traffic conditions** for selected road segments.

9. Emergency Route Support

The predicted traffic information is combined with the road network and available route options to support **time-efficient route decisions for ambulances, fire brigades and police vehicles**.

10. Communication-Cost Evaluation & Optimization

Throughout the federated process, we measure **model-update size, communication rounds and total data transferred**. We then evaluate optimization techniques such as reducing unnecessary rounds, update compression and selective participation to reduce communication cost while maintaining prediction performance.



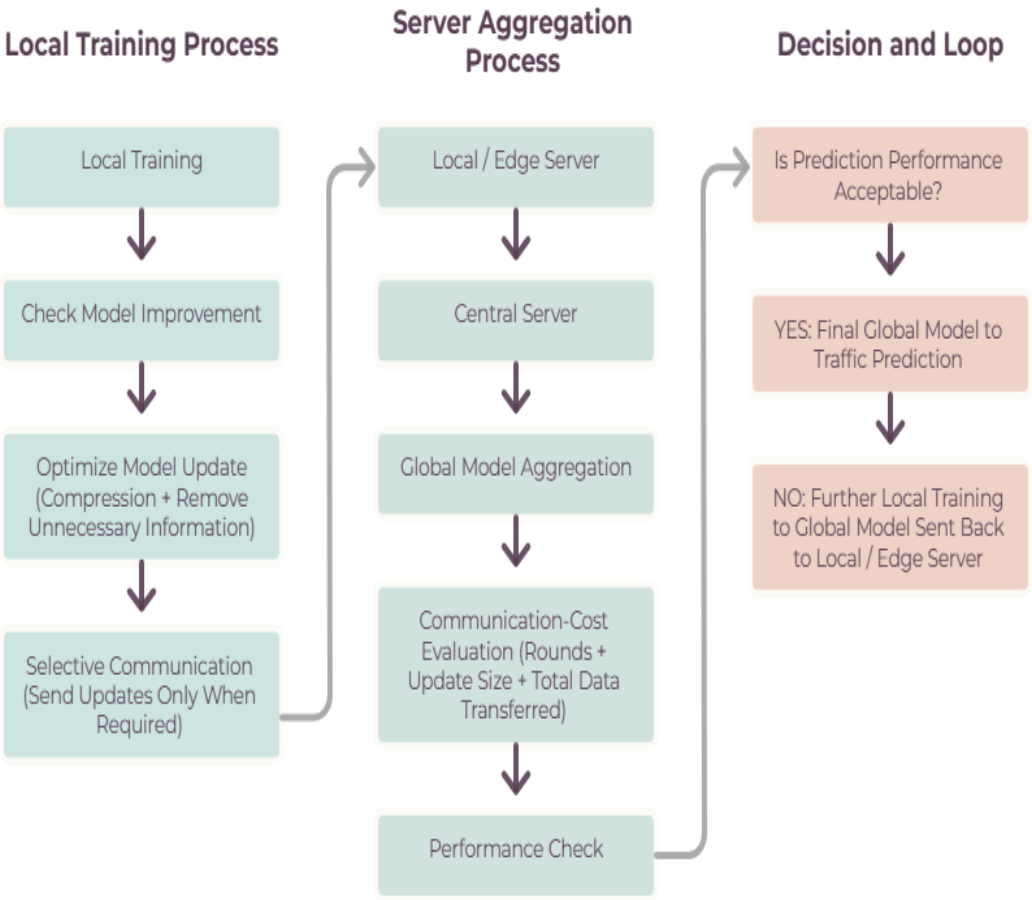
LIST OF OPTIMIZATIONS AND ENHANCEMENTS

1. **Optimized Communication Rounds** – Reduce unnecessary communication between local/edge servers and the central server.
2. **Model Update Compression** – Compress model updates before transmission to reduce the amount of data transferred.
3. **Selective Participation** – Allow only relevant or available locations to participate in a particular training round.
4. **Increased Local Training** – Perform more useful local training before sending an update, reducing the frequency of communication.
5. **Early Stopping** – Stop further communication when additional training rounds provide only negligible improvement.
6. **Communication-Cost Monitoring** – Track model size, number of rounds, and total data transferred during Federated Learning.
7. **Prediction–Communication Trade-off Analysis** – Evaluate whether reduced communication still maintains acceptable traffic-prediction performance.
8. **Emergency Route Integration** – Use the final traffic predictions as an input to an emergency-route decision-support module.

Table to compare before and after scenario

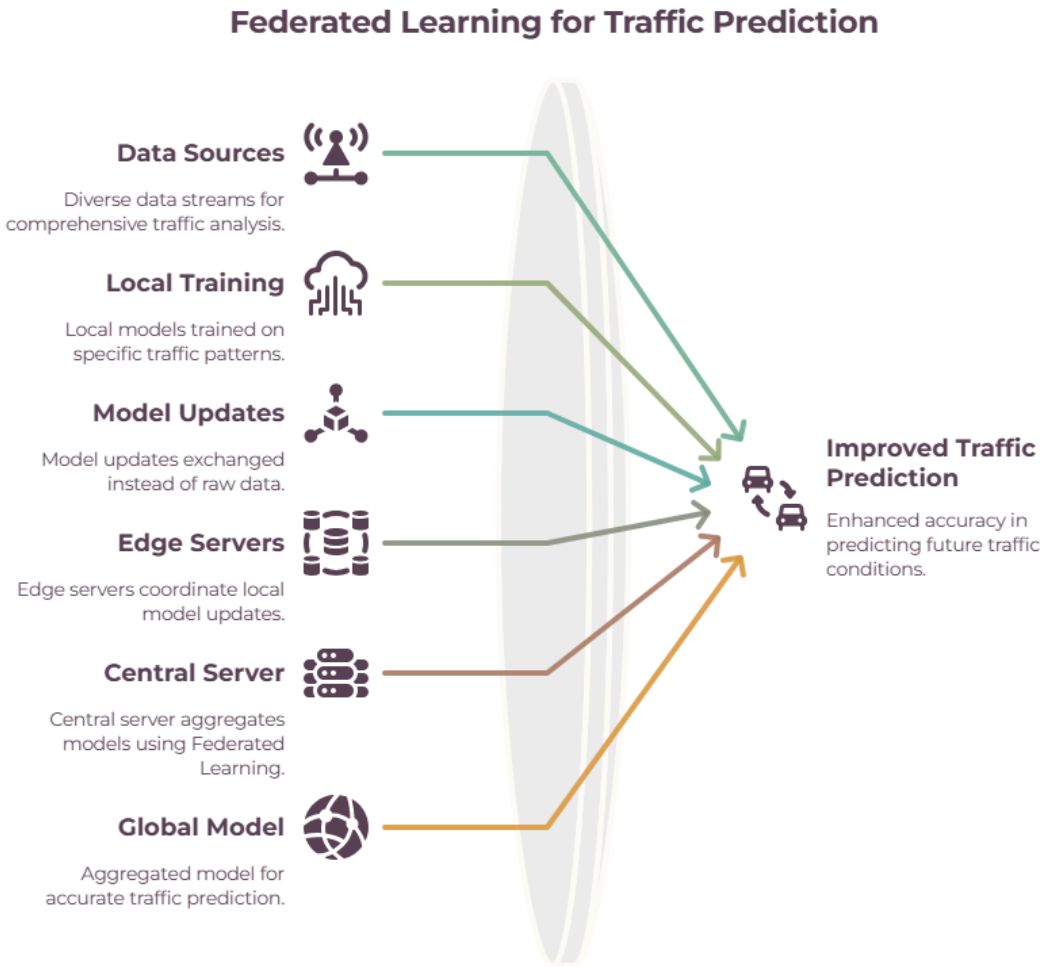
OPTIMIZATION	BEFORE	AFTER
Communication Rounds	Frequent model updates	Controlled communication rounds
Model Updates	Full-size updates transmitted	Compressed model updates
Location Participation	All locations communicate	Selective participation
Local Training	Less local training per round	More useful local training
Data Transfer	Higher communication volume	Reduced communication volume
Performance Monitoring	Focus mainly on accuracy	Accuracy + communication cost

OPTIMIZATION WORKFLOW

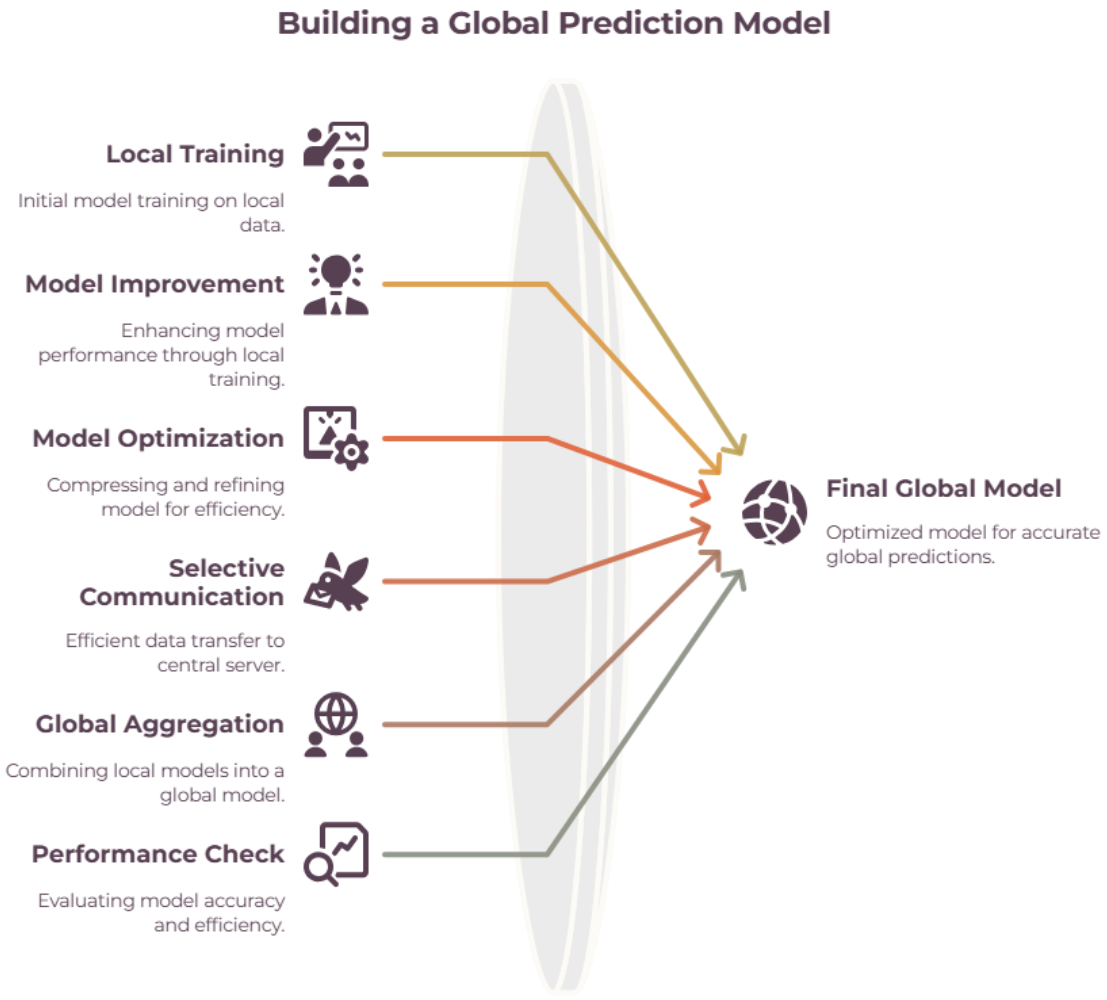


SOLUTION ARCHITECTURE AND DESIGN

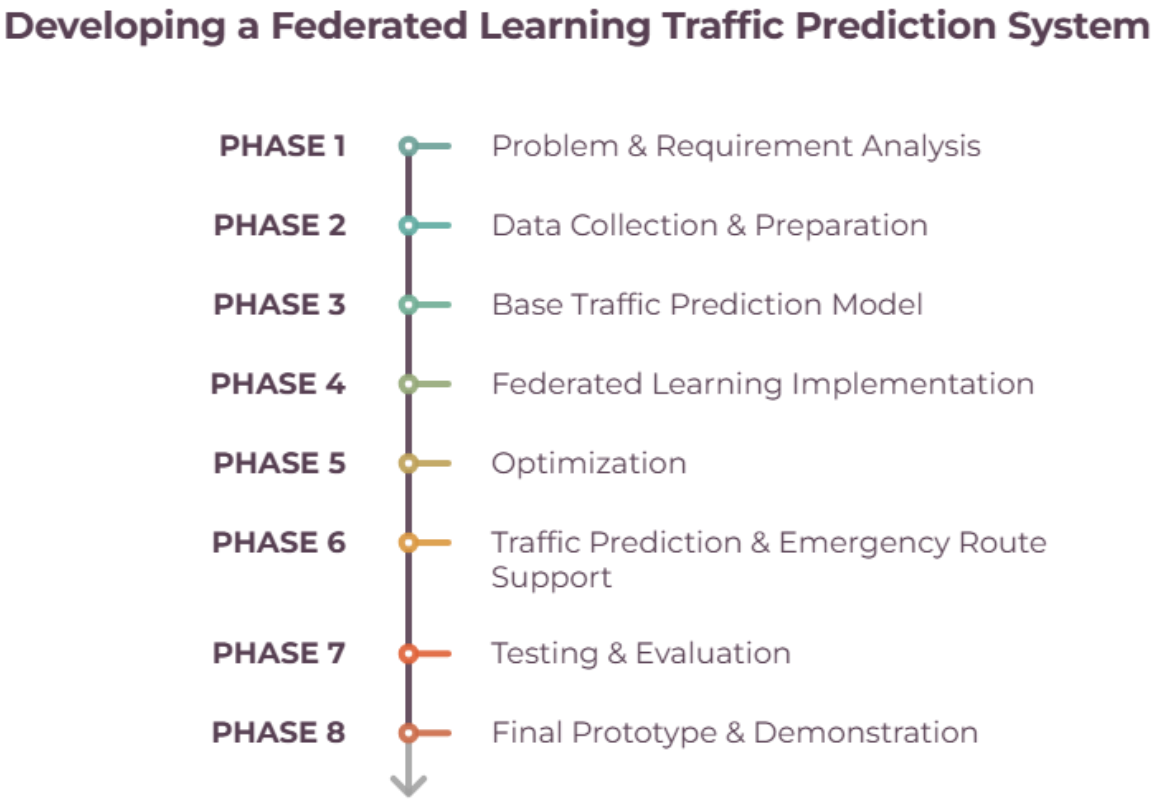
ARCHITECTURE WORKFLOW



OPTIMIZATION WORKFLOW



TIMELINE OF DELIVERY



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CONCLUSION

The proposed system combines multi-source traffic data, Federated Learning, communication optimization and emergency-route support into a unified traffic-management framework.

• Traffic data remains at the local level while model updates are shared for global learning.

• Repeated local and global training enables continuous improvement of the traffic-prediction model.

• Communication-efficient techniques help reduce unnecessary data transfer between local/edge servers and the central server.

• Predicted traffic conditions can support more informed route decisions for ambulances, fire brigades and police vehicles.

• The architecture is designed for the Chandigarh–Zirakpur corridor and can be extended to other road networks.

Overall, the solution aims to balance accurate traffic prediction, communication efficiency and effective emergency-route decision support.